

Troubleshooting Guide: Heat Exchanger — Tube Leak Causing Process Cross-Contamination

Category: Heat Transfer Equipment / Mechanical Integrity **Unit:** Shell-and-Tube Exchanger, Process-to-Process Service (Feed/Effluent Exchanger) **Tools:** Contaminant tracer/mass-balance trending, shell-vs-tube differential pressure comparison, eddy current tube inspection **Fluid Package:** PR for stream property context; the actual detection/diagnosis is a mechanical integrity investigation, not a phase-equilibrium problem **Symptom:** A trace contaminant appearing in a downstream product stream that has no other plausible source

1. Quick Reference

Item	Detail
Reported symptom	Trace contaminant detected in a downstream product stream with no obvious upstream source
Initially unclear	Whether this was a process/blending issue, an analyzer fault, or physical cross-contamination through a shared piece of equipment
Actual root cause	A tube-to-tubesheet joint failure (thermal cycling fatigue) in a feed/effluent exchanger allowed the higher-pressure side to leak into the lower-pressure side
Fix	Isolated the exchanger, plugged the leaking tube, scheduled a full retube at the next turnaround
Diagnostic signal	Contaminant concentration correlated specifically with a shift in the exchanger's shell-side/tube-side differential pressure, not with any upstream blending or analyzer calibration issue
Prevention	Periodic eddy current tube inspection tied to thermal cycling severity, not a fixed calendar interval

2. Symptom

- **A trace contaminant appeared in a downstream product stream** where it had no obvious plausible source — not present in the nominal feed, not explained by any known blending stream.

3. Why This Wasn't Assumed to Be an Analyzer or Blending Issue

An unexpected contaminant reading is often first suspected to be an **analyzer calibration issue** (a false reading) or a **blending/valve lineup error** (real contamination from a known but misrouted source). Both are more common and easier to check than a mechanical equipment

failure, so they were ruled out first — but a shared piece of equipment carrying two different process streams at different pressures, like a feed/effluent exchanger, is also a classic (if less obvious) cross-contamination pathway once those simpler explanations are eliminated.

4. Diagnostic Approach

Step 1 — Rule out analyzer error

The online analyzer was cross-checked against an independent lab sample, confirming the contaminant reading was **real**, not an instrument artifact.

Step 2 — Rule out blending/routing error

Valve lineups and blending stream sources were reviewed and confirmed correct — no known contaminant-bearing stream was being routed into the affected line.

Step 3 — Look for a shared-equipment pathway

With the simpler explanations ruled out, the process was mapped for **shared equipment** carrying both a stream that could plausibly contain the contaminant and the affected downstream stream — identifying a specific **feed/effluent exchanger** as a candidate, since its high-pressure side carried a stream containing the contaminant.

Step 4 — Confirm via differential pressure trend and physical testing

The exchanger's **shell-side and tube-side pressures** were reviewed, and a **differential pressure shift** was found correlating with the timing of the first contaminant detection. A hydrotest/tube inspection was performed to confirm the mechanism directly.

Quantitative Basis

- Contaminant concentration in the downstream product: normally <1 ppm (analyzer noise floor); detected at **14 ppm**, confirmed by independent lab sample at **13.5 ppm**.
- Exchanger design differential pressure (high-side minus low-side): 180 psi. Trended actual differential **dropped to 145 psi** over the same period the contaminant appeared — consistent with high-pressure fluid finding a leak path to the low-pressure side (reducing the net differential as some flow bypasses through the leak).
- Eddy current inspection identified **one tube with a through-wall indication at the tubesheet joint**, consistent with thermal cycling fatigue at that specific mechanical interface.

5. Root Cause

A tube-to-tubesheet joint failure, caused by thermal cycling fatigue, developed a through-wall leak path in the feed/effluent exchanger. This allowed a small amount of the high-pressure side stream to leak into the low-pressure side, carrying the contaminant downstream into the product stream — a mechanical integrity failure producing a symptom (unexpected contamination) that initially looked like a process or analytical issue.

6. Corrective Action

1. **Isolated the exchanger** from service.
2. **Plugged the leaking tube** to allow temporary continued operation of the exchanger (at slightly reduced heat transfer area).

3. **Scheduled a full retube** of the exchanger at the next planned turnaround.

7. Verification

- Downstream contaminant concentration returned to **<1 ppm** (below the analyzer noise floor) within hours of isolating and plugging the leaking tube.
- Exchanger differential pressure returned to **178 psi**, close to the 180 psi design value, confirming the leak path was closed.
- No further contaminant detections over the following **45 days** of continued operation on the plugged configuration.

8. Prevention / Long-Term Fix

- Established **periodic eddy current tube inspection tied to thermal cycling severity** (frequency and magnitude of start/stop or rate-change events this exchanger experiences), rather than a fixed calendar interval that might not reflect this specific exchanger's actual fatigue exposure.
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9. General Troubleshooting Checklist (Reusable)

- ☐ Cross-check any unexpected contaminant reading against an independent lab sample to rule out analyzer error first
- ☐ Review valve lineups and blending sources to rule out a known but misrouted contamination path
- ☐ If both are ruled out, map the process for **shared equipment** (exchangers, common headers) that could provide a physical pathway between a contaminant-bearing stream and the affected stream
- ☐ Review differential pressure trend across candidate shared equipment for a shift correlating with the contamination timing
- ☐ Confirm the mechanism directly via hydrotest, eddy current inspection, or equivalent
- ☐ Isolate/plug the identified leak path and confirm contaminant concentration and differential pressure both recover
- ☐ Schedule permanent repair (retube) at the next appropriate opportunity, since a plugged tube is a temporary measure
- ☐ Tie future inspection frequency to the specific exchanger's thermal cycling exposure, not a generic fixed interval

10. Key Takeaway

An unexplained contaminant in a product stream isn't always a process or analytical problem — it can be physical cross-contamination through a shared piece of equipment like a feed/effluent exchanger, especially once the more common explanations (analyzer error, blending error) are ruled out. A shift in an exchanger's shell-to-tube differential pressure, reviewed alongside the contamination timing, can point directly at a developing tube leak well before it becomes large enough to show up as an obvious mechanical failure.

Related Concepts / Tags

heat-exchanger tube-leak cross-contamination tubesheet-joint thermal-cycling-fatigue
eddy-current-inspection mechanical-integrity differential-pressure shell-and-tube

This guide is derived from a representative field troubleshooting scenario. Values and thresholds shown are illustrative and should be validated against your own unit's design basis before applying.